3.0 BHS-SLQ LOWER DECK BRIDGE DESCRIPTION

The existing bridge is highly corroded and its strengthening is not feasible. Hence New bridge (Walkway, power cables, piping etc.) is envisaged between BHS & SLQ platforms.
The landing point at the SLQ platform is identified near grid B2 (southside). The landing point at the BHS platform is identified near grid A1 (southside).



Figure 1: BRIDGE (existing and proposed location-BHS SIDE)

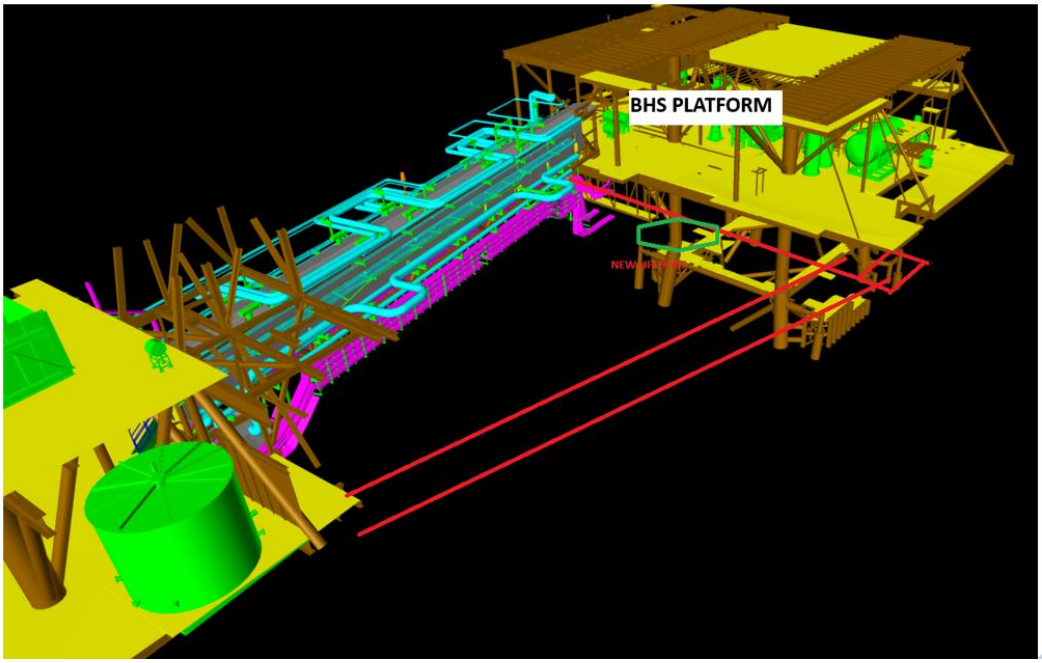


Figure 2: BRIDGE (existing and proposed locations)

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Approximate length of bridge is measured from 3D model as 40m. Hence, 42m length of bridge is modelled in SACS conservatively.

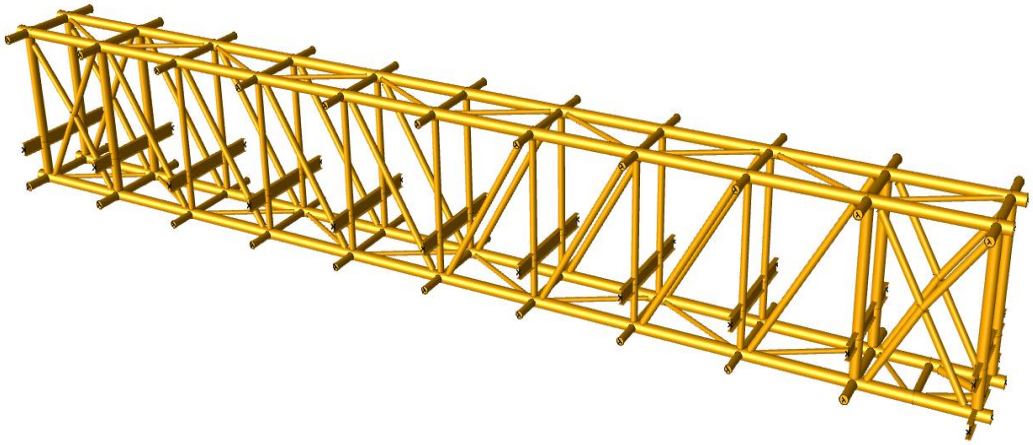


Figure 3: 3D SACS MODEL

Length of Bridge	42 m
Support Condition	Fixed: 001000, 111000, 001000 (SLQ Side)) Sliding: 011000, 011000 (BHS Side)
No. of directions for environmental loads	8
Wind Speed (3 Sec Gust)	65.45 m/s for extreme and 39.99 m/s for Operating.
Density of steel	7.850 T/m3
Density of water	1.025 T /m3
Yield strength of steel	50ksi
Environmental parameters	As per design basis (DC-3.4-Part-1-Rev.14)

ISOMETRIC VIEW OF **BHS-SLQ Lower Deck Bridge** is shown in Figure1-3.

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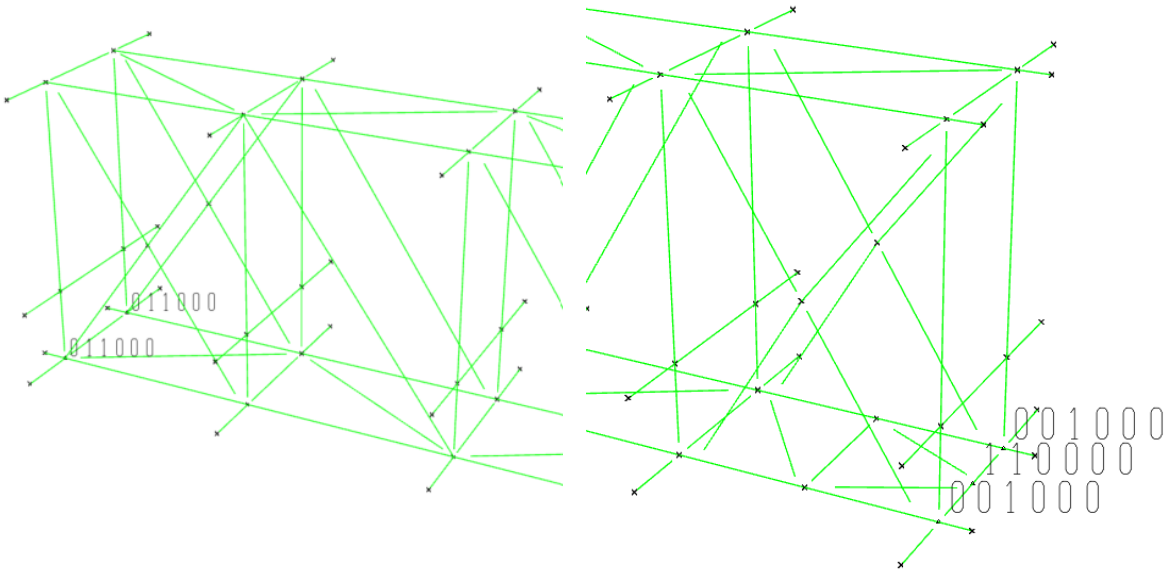


Figure 4: SUPPORT CONDITIONS

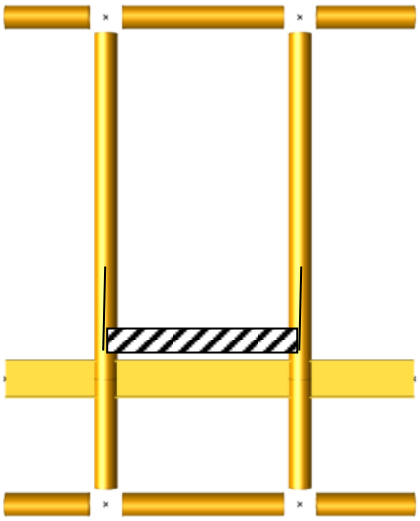


Figure 5: BRIDGE CROSS SECTION

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4.0 LOADING

Basic Load Cases

LOADCN	DESCRIPTION	LOAD (kN)
DL	Self-weight, (Contingency - 13%), Walkway Member Wt	789.79
NGDL	Bracket load	266.47
	Extension	
	Grating Load	
	Grating support load	
	Handrail Load	
	Handrail support load	
	Light support	
	Monorail support	
	Roof Clad	
	Sheeting support	
	Tray support	
LL	Blanket live load on Walkway	268.42
MRLL	Monorail Live Load	20.00
CL	Cable Loading	77.66
PLEM	Piping Load Empty (60% of PLOP)	864.00
PLOP	Piping Load Empty + Operating Contents (By Piping)	1440
PLTLX	Piping Load Empty + Operating Contents (30% of	432.16
PLTLY	~~Piping L~~oad Empty + Operating Contents (30% of	432.16
201 to 208	Extreme Wind load	See Annexure
301 to 308	Operating Wind Load	See Annexure
FLX	Bridge Friction Load X (30 % of SW+NGDL+LL+CL+PLOP)	926
FLY	Bridge Friction Load Y (30 % of SW+NGDL+LL+CL+PLOP)	926

SELF WEIGHT

Members are modelled and the self-weight is considered in the model. Contingency (13 %) on the modelled structural steel has been captured by increasing the density of the modelled members.

NON-GENERATED DEAD LOAD

The weight of members which are not modelled in SACS is considered as NGDL. This includes the Cable tray, cable tray support, grating and handrail loadings.

LIVE LOAD

Blanket live load (grating area live load) on the walkway is considered as 250kg/m2. 2.324 m width of walkway is considered.

CABLE TRAY AND CABLE LOAD

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3.0 ICW-ICG LOWER DECK BRIDGE DESCRIPTION

The existing bridge has fallen during the cyclone. Hence New bridge (Walkway) is envisaged between ICG & ICW platforms.
The landing point at the ICW platform is identified near grid 4 (northside near loading unloading area). The landing point at the ICG platform is identified near grid 3 (southside).
Approximate length of bridge is measured from 3D model as 54.5m. Hence, 56m length of bridge is modelled in SACS conservatively.
No piping is envisaged in the existing bridge. Only cable loading is considered for Lighting purpose.

Length of Bridge	56 m
Support Condition	Fixed: 001000, 111000, 001000 (ICG Side)) Sliding: 011000, 011000 (ICW Side)
No. of directions for environmental loads	8
Wind Speed (15 Sec Gust)	61.11 m/s for extreme and 37.7 m/s for Operating.
Density of steel	7.850 T/m3
Density of water	1.025 T /m3
Yield strength of steel	50ksi
Environmental parameters	As per design basis (DC-3.4-Part-1-Rev.14)

ISOMETRIC VIEW OF **ICG-ICW Lower Deck Bridge** is shown in Figure1.

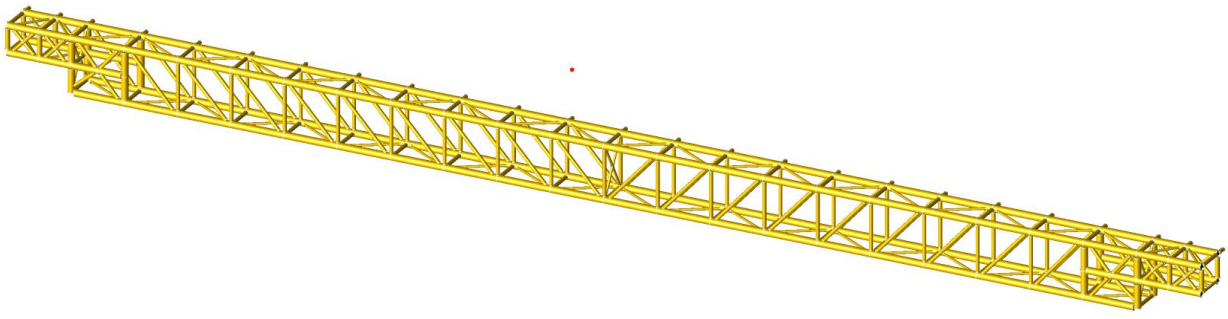


Figure 1: 3D SACS MODEL

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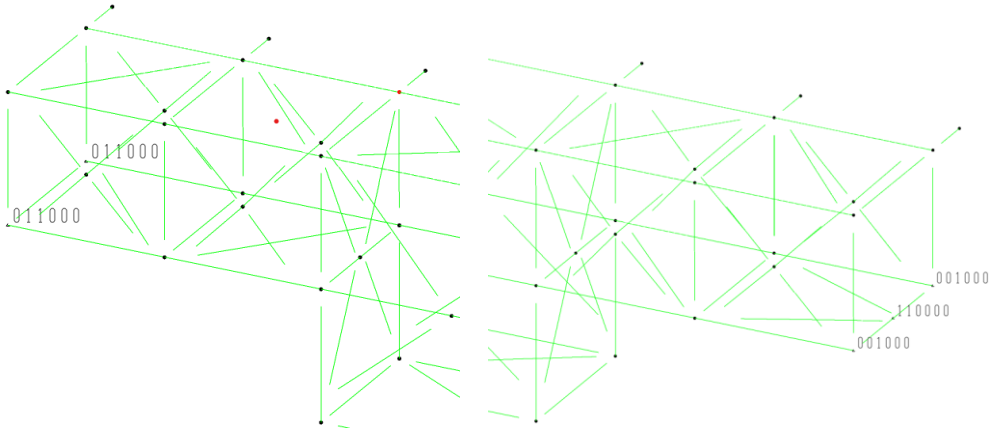


Figure 2: SUPPORT CONDITIONS

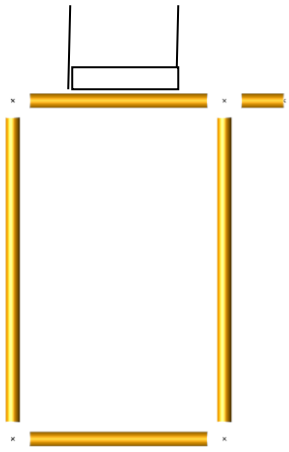


Figure 3: BRIDGE CROSS SECTION

4.0 LOADING

Basic Load Cases

LOADCN	DESCRIPTION	LOAD (kN)
DL	Self-weight, (Contingency - 13%), Walkway Member Wt	418.03
NGDL	Cable tray	155.75
	Cable tray Support	
	Grating Load	
	Handrail Load	

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LL	Blanket live load on Walkway	156.25
CL	Cable Loading	0.750
201 to 208	Extreme Wind load	See Annexure
301 to 308	Operating Wind Load	See Annexure
FLX	Bridge Friction Load X (30 % of SW+NGDL+LL)	216
FLY	Bridge Friction Load Y (30 % of SW+NGDL+LL)	216

SELF WEIGHT

Members are modelled and the self-weight is considered in the model. Contingency (13 %) on the modelled structural steel has been captured by increasing the density of the modelled members.

NON-GENERATED DEAD LOAD

The weight of members which are not modelled in SACS is considered as NGDL. This includes the Cable tray, cable tray support, grating and handrail loadings. .

LIVE LOAD

Blanket live load (grating area live load) on the walkway is considered as 250kg/m2. 1.2 m width of walkway is considered.

CABLE TRAY AND CABLE LOAD

For 450mm cable tray, 21.85kg/3m load is considered. Cable trays consist of 4 cables (2no.s 3C x 2.5 m2 with load 162kg/km, 2no.s 4C x 16 m2 with load 350kg/km)

WIND LOAD

Wind load approaching the bridge are considered in 8 directions at 45-degree sectors for both 1-year operating and 100-year extreme storm in the structural analysis. Wind load in each direction is calculated based on projected areas with the applicable shape factor (Cs).

For all the load cases, the detailed calculation is provided in ANNEXURE-1.

4.1 Load Combinations

Load Combinations

BASIC LOAD COMBINATIONS										
LCOMB	Load Case	Factor	Load Case	Factor	Load Case	Factor	Load Case	Factor	Load Case	Factor
1000	DL	1.00	NDGL	1.13	CL	1.00				
1001	DL	1.00	NDGL	1.13	CL	1.00				
1002	DL	1.00	NDGL	1.13	CL	1.00				
1003	DL	1.00	NDGL	1.13	CL	1.00				
1004	DL	1.00	NDGL	1.13	CL	1.00	LL	1.00		
1005	DL	1.00	NDGL	1.13	CL	1.00	LL	1.00		

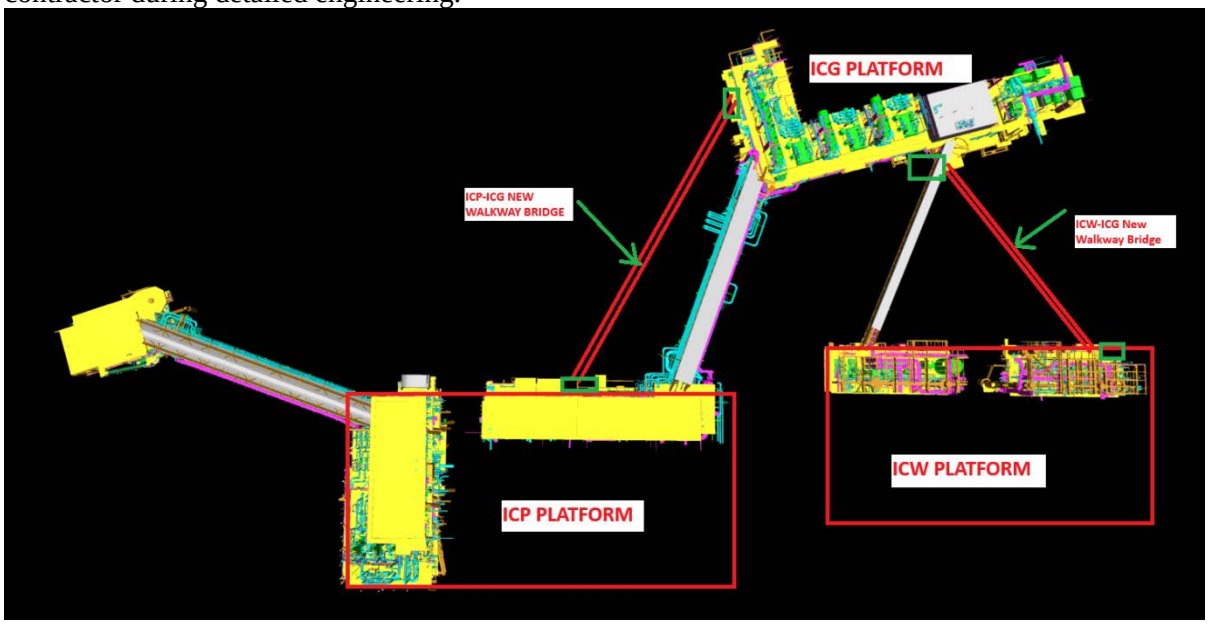
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3.0 ICP-ICG LOWER DECK BRIDGE DESCRIPTION

The existing bridge has fallen during the cyclone. Hence New bridge (Walkway) is envisaged between ICG & ICP platforms.

The landing point at the ICP platform is identified near grid 3 (northside between loading area and Survival craft). The landing point at the ICG platform is identified near the Scramble Net area (Between staircase and landing Area).

Approximate length of bridge is measured from 3D model as 80.5m. Hence, 84m length of bridge is modelled in SACS conservatively. The exact length of the bridge shall be measured by contractor during detailed engineering.



It was suggested from ICP platform site that the ICP new landing location shall be considered at the present lifeboat location. (Shifting the life boat westwards). This shall reduce the gap between the existing ICP-ICG Upper deck bridge and new Bridge. (Approximate gap of 3 m is only available presently). The same is huge risk during installation of new bridge (considering projection of existing live piping from the ICP-CIG UD bridge). Hence identified landing location (identified during the combined site visit) is retained.

No piping is envisaged in the existing bridge. Only cable loading is considered for Lighting purpose.

Length of Bridge	84 m
Support Condition	Fixed: 001000, 111000, 001000 (ICG Side)) Sliding: 011000, 011000 (ICP Side)
No. of directions for environmental loads	8
Wind Speed (15 Sec Gust)	57.19 m/s for extreme and 35.68 m/s for Operating.
Density of steel	7.850 T/m3
Density of water	1.025 T /m3

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Yield strength of steel	50ksi
Environmental parameters	As per design basis (DC-3.4-Part-1-Rev.14)

ISOMETRIC VIEW OF **ICG-ICP Lower Deck Bridge** is shown in Figure1.

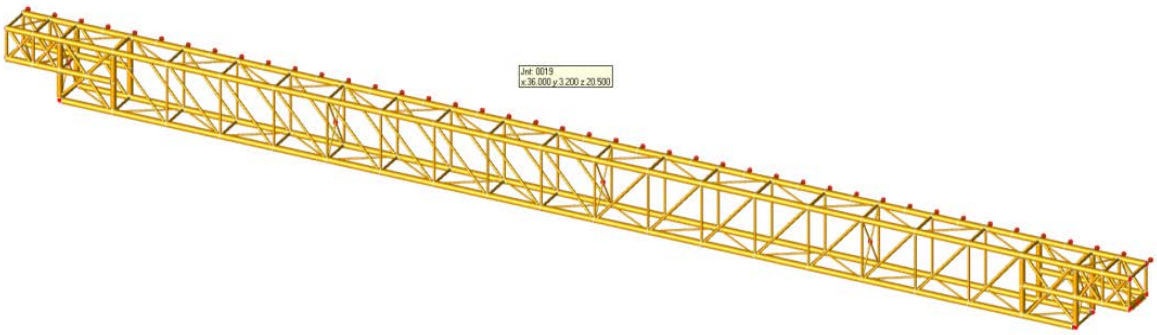


Figure 1: 3D SACS MODEL

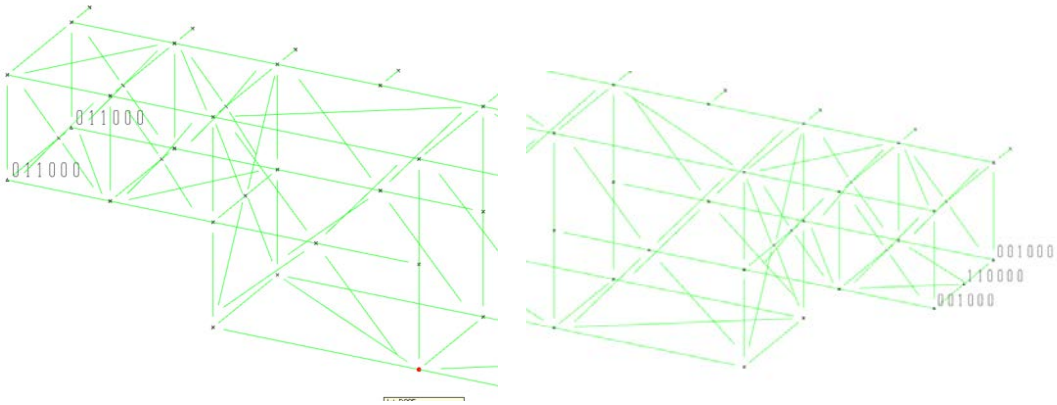


Figure 2: SUPPORT CONDITIONS

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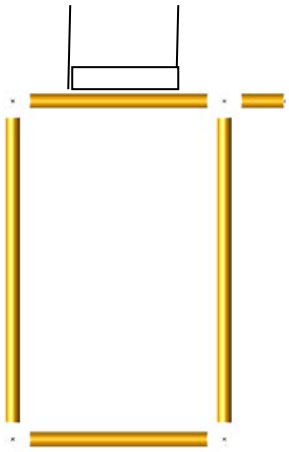


Figure 3: BRIDGE CROSS SECTION

4.0 LOADING

Basic Load Cases

LOADCN	DESCRIPTION	LOAD (kN)
DL	Self-weight, (Contingency - 13%), Walkway Member Wt	895.54 +71.1
NGDL	Cable tray Support	3.13
	Grating Load	57.6
	Handrail Load	96.0
LL	Blanket live load on Walkway	288
MRLL	Monorail Live Load	Nil
CL	Cable Loading	0.882
201 to 208	Extreme Wind load	See Annexure
301 to 308	Operating Wind Load	See Annexure
FLX	Bridge Friction Load X (30 % of SW+NGDL+LL+CL+PLOP)	342
FLY	Bridge Friction Load Y (30 % of SW+NGDL+LL+CL+PLOP)	342

SELF WEIGHT

Members are modelled and the self-weight is considered in the model. Contingency (13 %) on the modelled structural steel has been captured by increasing the density of the modelled members.

NON-GENERATED DEAD LOAD

The weight of members which are not modelled in SACS is considered as NGDL. This includes the Cable tray, cable tray support, grating and handrail loadings. .

LIVE LOAD

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Blanket live load (grating area live load) on the walkway is considered as 250kg/m2. 1.2 m width of walkway is considered.

CABLE TRAY AND CABLE LOAD

For 450mm cable tray, 21.85kg/3m load is considered. Cable trays consist of 4 cables (2no.s 3C x 2.5 m2 with load 162kg/km, 2no.s 4C x 16 m2 with load 350kg/km)

WIND LOAD

Wind load approaching the bridge are considered in 8 directions at 45-degree sectors for both 1-year operating and 100-year extreme storm in the structural analysis. Wind load in each direction is calculated based on projected areas with the applicable shape factor (Cs).

For all the load cases, the detailed calculation is provided in ANNEXURE-1.

4.1 Load Combinations

Load Combinations

BASIC LOAD COMBINATIONS										
LCOMB	Load Case	Factor	Load Case	Factor	Load Case	Factor	Load Case	Factor	Load Case	Factor
1000	DL	1.00	NDGL	1.13	CL	1.00				
BASIC LOAD COMBINATIONS WITH BRIDGE FRICTION										
1011	1000	1.00	FLX	1.00						
1012	1000	1.00	FLX	-1.00						
1013	1000	1.00	FLY	1.00						
1014	1000	1.00	FLY	-1.00						
LOAD COMBINATIONS FOR EXTREME STORM ENVIRONMENT										
2001 to 2038	1011 to 1014	1.00	201 to 208	1.00						
LOAD COMBINATIONS FOR OPERATING STORM ENVIRONMENT										
3001 to 3038	1011 to 1014	1.00	301 to 308	1.00						

5.0 RESULTS & SUMMARY

BASIC LOAD CASE SUMMARY SACS OUTPUT

****** SEASTATE BASIC LOAD CASE SUMMARY ******

RELATIVE TO MUDLINE ELEVATION

MARINE METHOD

LOAD CASE	LOAD LABEL	FX (KN)	FY (KN)	FZ (KN)	MX (KN-M)	MY (KN-M)	MZ (KN-M)	DEAD LOAD (KN)	BUOYANCY (KN)

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